Training and extracting estimates from KenLM

Sources:
- Code: https://github.com/kpu/kenlm
- Documentation: https://kheafield.com/code/kenlm/

There are three main steps:

1. Installation
2. Training the n-gram model
3. Extracting estimates

Installation:

a. Install dependencies depending on your system: kenlm/BUILDING at master · kpu/kenlm · GitHub
b. Clone/download the repo to your computer
c. Compile KenLM: kpu/kenlm: KenLM: Faster and Smaller Language Model Queries (github.com)
d. Install the python package (kenlm/README.md at master · kpu/kenlm · GitHub) with *pip install https://github.com/kpu/kenlm/archive/master.zip*

Training the n-gram model:

From the code folder run:

bin/lmplz -o 5 <text >text.arpa

- Might need to change *bin/lmplz* to *build/bin/lmplz*
- See estimation . kenlm . code . Kenneth Heafield (kheafield.com) for parameter options (such as choosing the `n`) and for corpus formatting notes.
- Replace 'text' with the input text file.
- The resulting model will be stored as `text.arpa`.

Extracting estimates:

Can use command line or python.
In Python:

- Load a pretrained model with `import kenlm; model = kenlm.Model('<model_path>')

- Extract probabilities with the `model.full_scores` function. See example usage here - https://github.com/kpu/kenlm/blob/5bf7b46558e1c5595bf3b8c9b0b1f9d8d257040a/python/example.py#L20-L24
- Probabilities are log10 probabilities see scroe function documentation for more details. https://github.com/kpu/kenlm/blob/5bf7b46558e1c5595bf3b8c9b0b1f9d8d257040a/python/kenlm.pyx#L153
- Full python example: https://github.com/kpu/kenlm/blob/master/python/example.py
- Full python function documentation: kenlm/python/kenlm.pyx at master · kpu/kenlm · GitHub